"Gamified Cyber Education: Measuring Behavior Change"

Bachelor Thesis — Project Plan / Strategy

Submitted at the Faculty of Management Technology

German International University

Submitted by:

Eyad Saeed

Student ID:

13005578

Supervisor:

Prof. Dr.-Ing. Maggie Mashaly

Research Theme:

Play, Learn & Protect — Technology for Childhood

Date:

Sunday, September 20th, 2026

1. Topic Articulation, Research Question and Motivation

Children today are frequent users of digital platforms — social media, gaming, messaging apps — yet most of them have little awareness of the risks they face online, such as phishing attempts, unsafe sharing of personal information, or weak password habits. Existing digital literacy programs often address this through traditional instruction: slides, videos, and rule-based lessons. The well-documented problem with this approach is that children can learn the rules without actually changing how they behave online afterward.

This project addresses that gap by designing a short gamified cyber-awareness module specifically aimed at school children, built using a no-code interactive platform, drawing on established gamification frameworks applied in cybersecurity education research — notably Cyber-Hero (Qusa & Tarazi, 2021) and CyberKids — implemented through tools such as Genially or H5P which support scenario-based modules without custom software development. The module teaches one specific online safety behavior through game elements — levels, points, instant feedback, and rewards — and the project then measures whether school children's actual behavior changes after completing it, compared to before, using a pre-test/post-test design.

The three behavior areas described in the topic are password hygiene, phishing recognition, and oversharing awareness. Given that the target audience is school children, oversharing awareness — understanding what personal information is safe to share online — is a particularly fitting focus, as it is both age-relevant and measurable through scenario-based questions.

Proposed research question: To what extent does a short gamified cyber-awareness module produce a measurable change in school children's online safety behavior, as measured by a pre-test/post-test design?

This project sits within the Play, Learn & Protect research theme, which centers on protecting children in digital environments. The target population is school children, and all data collection will be subject to appropriate ethical review and parental consent procedures.

2. Early Literature Findings

An initial literature search was carried out focused specifically on gamified cybersecurity and online safety education for school children, existing approaches to measuring behavior change in this population, and the design considerations involved in working with school children.

2.1 The gap between knowledge and behavior in school children's online safety

Quayyum, Cruzes and Jaccheri (2021) published a systematic literature review mapping existing cybersecurity awareness tools, methods, and evaluation approaches used specifically with school children. A key finding was that most existing programs focus on knowledge transfer — what children know about online risks — rather than measuring whether their actual online behavior changes as a result. This directly establishes the gap this project aims to address: designing and testing an intervention that targets behavior, not just awareness.

Akhter, Tabassum and Khatun (2020), in a survey on cybersecurity challenges and awareness for school children published through IEEE, found that children are particularly vulnerable to online threats precisely because of their limited awareness and tendency to trust digital interactions. The study highlighted that both school children and their parents often lack basic understanding of online safety behaviors, reinforcing the need for engaging, accessible education formats targeted at school-age audiences.

2.2 Evidence that gamified approaches work with school children

Katsantonis, Fidas, Mikropoulos and Chalki (2020) conducted a quasi-experimental study evaluating a gamified interactive learning environment designed to improve school children's e-safety skills. Using a pre-test/post-test design, they found that children who completed the gamified program reported significantly more positive and safe online behaviors at post-test compared to pre-test. This paper is the closest methodological match to this project — same target population (children), same design (pre/post), same domain (online safety) — and serves as a direct template for the methodology.

The peer-learning and game-based instruction study for primary school cybersecurity (Videnovik et al., 2024) found that combining game-based learning with structured knowledge activities produced long-lasting improvements in school children's cybersecurity knowledge. The study is relevant as it demonstrates that game-based approaches are not only

engaging but can also produce durable learning outcomes in school children when designed with the right structure.

2.3 Gamification and behavior change in educational settings

Kim and Castelli (2021) conducted a meta-analysis of gamified interventions in educational settings and their effects on student behavioral change. Across the studies reviewed, they found that gamified reward systems — including digital badges, points, and leaderboards — were consistently associated with improved student engagement and behavioral outcomes. While not specific to cybersecurity, this meta-analysis provides robust theoretical grounding for the use of gamification as a behavior-change mechanism in an educational context with school-age learners.

2.4 Measurement and ethical considerations for school children

A key methodological challenge for this project is the measurement instrument. The SeBIS scale (Egelman et al., 2016), while a validated and widely used tool for measuring security behavior, was designed and validated with adult populations. For school children, a simplified, scenario-based pre/post instrument is more appropriate — presenting children with realistic online situations and asking what they would do, rather than abstract self-report items. The instrument design draws on the measurement approaches used in Katsantonis et al. (2020) and the Quayyum et al. (2021) review, using scenario-based questions appropriate for school children.

Regarding data collection, working with school children requires parental consent, and ideally coordination through a school or community organization. Given that this project is part of the Play, Learn & Protect research theme, data collection will be conducted through school coordination, with parental consent obtained prior to any participation.

3. Proposed Timeline

The timeline below follows the 14-week workplan structure from the topic description, mapped onto the official GIU schedule dates (start: 12 September 2026; final submission: 5

January 2027). Data collection access and parental consent procedures are built into Phase 2, as these require lead time when working with school children.

Phase / Date	**Focus**	**Key Activities**
Phase 1 — Wks 1–2 (Sep 12 – Sep 25)	Scoping & Literature Review	Complete initial literature review; finalize gamified module scope and behavior focus (oversharing awareness); submit project plan by Sep 20.
Phase 2 — Wks 3–4 (Sep 26 – Oct 9)	Module & Instrument Design	Design gamified module content appropriate for school children; design pre/post test instrument adapted for the target age group; coordinate data collection access (school/parental consent).
Phase 3 — Wks 5–8 (Oct 10 – Nov 6)	Data Collection	Obtain parental consent; administer pre-test; children complete gamified module; administer post-test.
Interim Report 1 (Oct 13)	—	First interim report: scope confirmed, literature review completed, module design in progress, data collection access confirmed.
Phase 4 — Wks 9–11 (Nov 7 – Nov 27)	Analysis & Refinement	Run paired t-tests on pre/post data; interpret results; refine module if needed; validate with supervisor.
Interim Report 2 (Nov 15)	—	Second interim report: data collected, preliminary pre/post findings included.
Phase 5 — Wks 12–13	Write-up	Draft methodology, results, discussion, limitations, and conclusion chapters.
Phase 6 — Wk 14	Defense Preparation	Finalize slides; rehearse presentation; incorporate supervisor feedback.
Final Submission (Jan 5, 2027)	—	Submit 3 hard copies + digital PDF to GIU library (Thesis Regulations §5.4).

References

Akhter, F., Tabassum, N., & Khatun, R. (2020). A Survey on Cybersecurity Challenges and Awareness for Children of All Ages. In 2020 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE). IEEE. https://doi.org/10.1109/CSDE50874.2020.9231229

Videnovik, M., Trajkovik, V., Vold, T., Kiønig, L. V., Madevska Bogdanova, A., & Filiposka, S. (2024). Using Peer-Learning and Game-Based Instruction for Achieving Long-Lasting Knowledge of Cybersecurity in Primary Schools. IEEE Access, 13, 11679–11688. https://doi.org/10.1109/ACCESS.2024.3479921

Katsantonis, M., Fidas, C., Mikropoulos, T. A., & Chalki, P. (2020). Improving Children's E-Safety Skills through an Interactive Learning Environment: A Quasi-Experimental Study. Multimodal Technologies and Interaction, 4(2), 10. https://doi.org/10.3390/mti4020010

Kim, J., & Castelli, D. M. (2021). Effects of Gamification on Behavioral Change in Education: A Meta-Analysis. International Journal of Environmental Research and Public Health, 18(7), 3550. https://doi.org/10.3390/ijerph18073550

Quayyum, F., Cruzes, D. S., & Jaccheri, L. (2021). Cybersecurity awareness for children: A systematic literature review. International Journal of Child-Computer Interaction, 30, 100343. https://doi.org/10.1016/j.ijcci.2021.100343

Qusa, H., & Tarazi, J. (2021). Cyber-Hero: A Gamification Framework for Cyber Security Awareness for High Schools Students. In 2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC), pp. 677–682. IEEE. https://doi.org/10.1109/CCWC51732.2021.9375847

Pérez, J., Torres, R., & von Brand, S. (2020). CyberKids: A Video Game for Raising Cyber Security Awareness in Children. In 2020 39th International Conference of the Chilean Computer Science Society (SCCC). IEEE. https://doi.org/10.1109/SCCC51225.2020.9281253